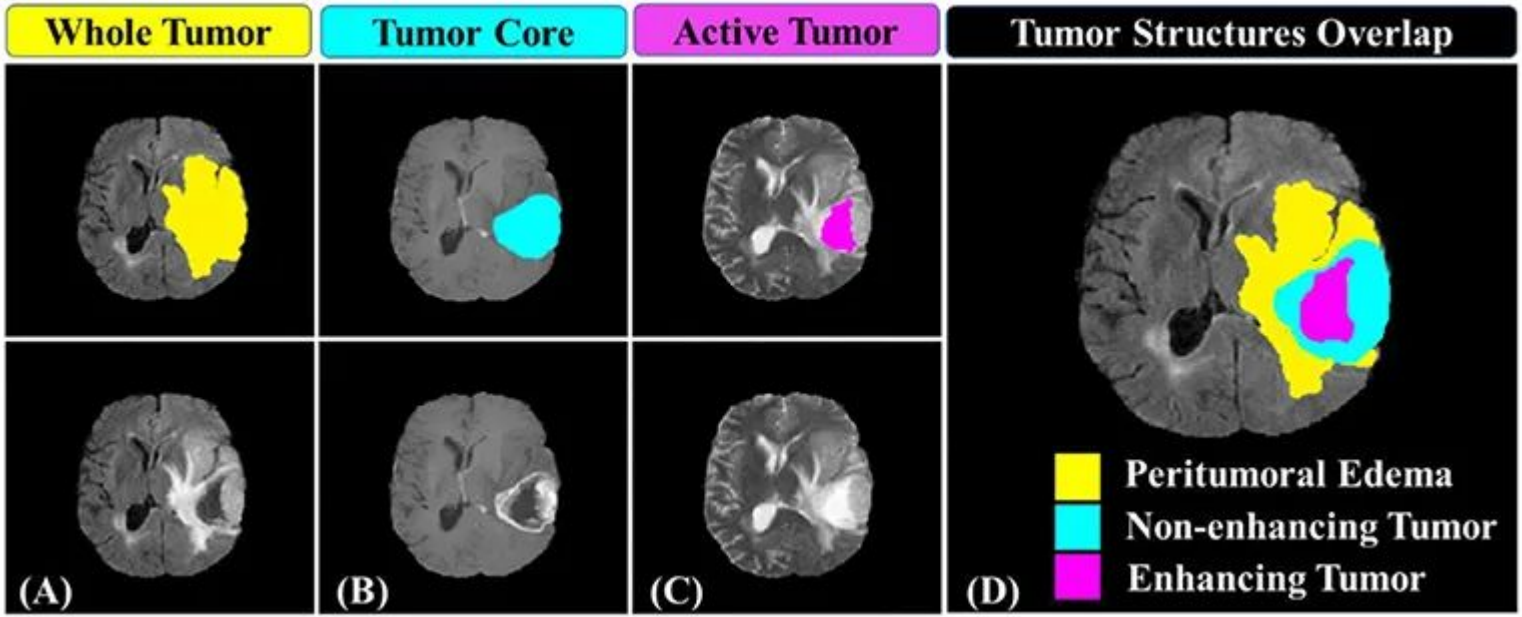
“Brain Tumor Segmentation with Synthetic MR Images”

GROUP MEMBERS
Saugat Bhattarai
Bijay Adhikari
Piyush Dawadi

DATE: Oct. 8, 2024

OBJECTIVE

Developing AI Model that accurately delineates tumor boundaries and sub-regions



STATEMENT OF PROBLEM

- To the point explanation without any filler statements and exaggeration.

Literature Review

3D U-Net for Brain Tumour Segmentation

Architecture: Enhanced 3D U-Net for BraTS 2018 dataset

Loss Function: Weighted Categorical Cross-Entropy (CCE)

Performance (Dice Scores):

- Enhancing Tumor (ET): 0.706
- Whole Tumor (WT): 0.871
- Tumor Core (TC): 0.771

Evaluation Metrics: Dice Score, Sensitivity, Specificity, Hausdorff-95 distance

Literature Review

Data Augmentation for Brain-Tumor Segmentation: A Review

Techniques

1. **Affine Transformations**: Rotation, flipping, scaling, cropping.
2. **Elastic Transformations**: Distorts images while preserving topology.
3. **Pixel-Level Transformations**: Modifies pixel intensities.
4. **Artificial Data Generation**: Techniques like GANs for synthetic data creation.

Evaluation Metrics

- **Primary Metric**: Dice Similarity Coefficient (DSC)
- **Other Metrics**: Sensitivity, Specificity, Hausdorff Distance

Literature Review

Using a generative adversarial network to generate synthetic MRI images for multi-class automatic segmentation of brain tumors

Methods

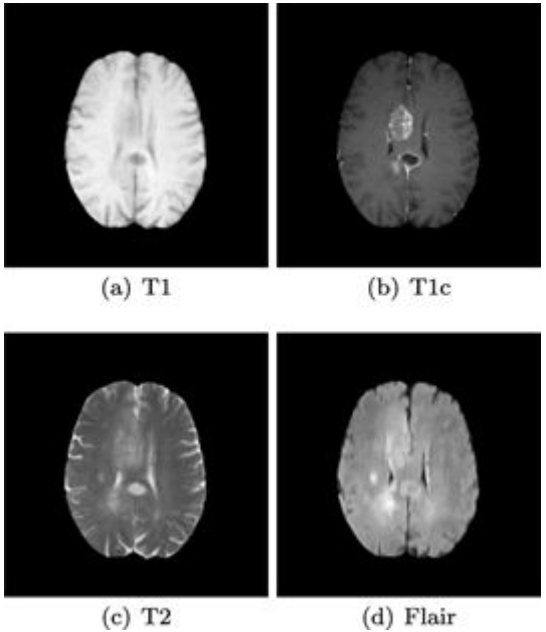
- **Pix2PixNIfTI GAN**: Generated synthetic MRI contrasts from available ones
- **DeepMedic (3D CNN)**: Used to evaluate segmentation performance

Evaluation Metrics: Dice coefficient, MSE

Results

- Four-input DeepMedic model (all original images) achieved highest Dice score for whole tumor (WT): 0.93 ± 0.06.
- Synthetic T2w images provided best performance among synthetic images, with WT Dice score: 0.90 ± 0.08.
- Three-input models like T1CE-T2w-FLAIR showed strong segmentation results, particularly for necrotic core (NC) and edema (ED)

Image Modalities



a)T1-weighted (T1)
- shows anatomy and structural detail
- useful for identifying normal brain tissue

b)T1 with gadolinium enhancing contrast (T1c)
- T1 with contrast enhancement
- highlights abnormal areas like tumors or inflammation

c)T2-weighted (T2)
- shows fluid and edema
- useful for detecting lesions and abnormal tissue

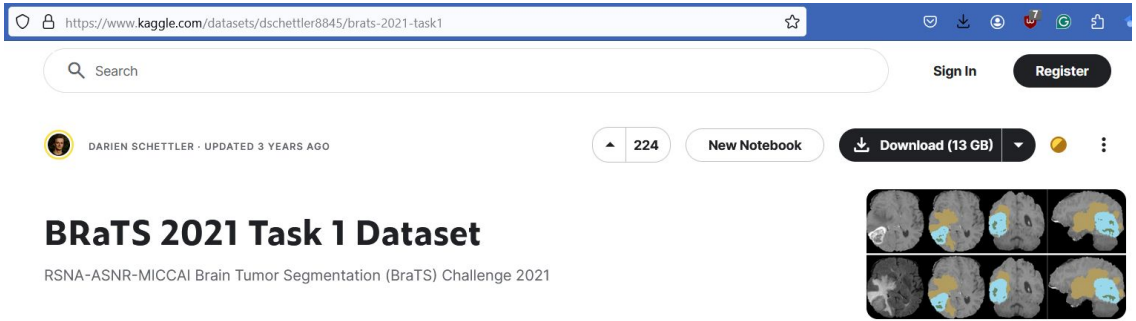
d)Fluid Attenuation Inversion Recovery (FLAIR)
- suppresses fluid signals
- highlights lesions near CSF, useful in identifying abnormalities like tumors or multiple sclerosis.

CONTRIBUTION AREAS

NAME	PROJECT CONTRIBUTION
Saugat Bhattarai	● Literature Review ● Image Modalities ● Data Augmentation

ABOUT DATASETS

BraTS2021 dataset available at kaggle

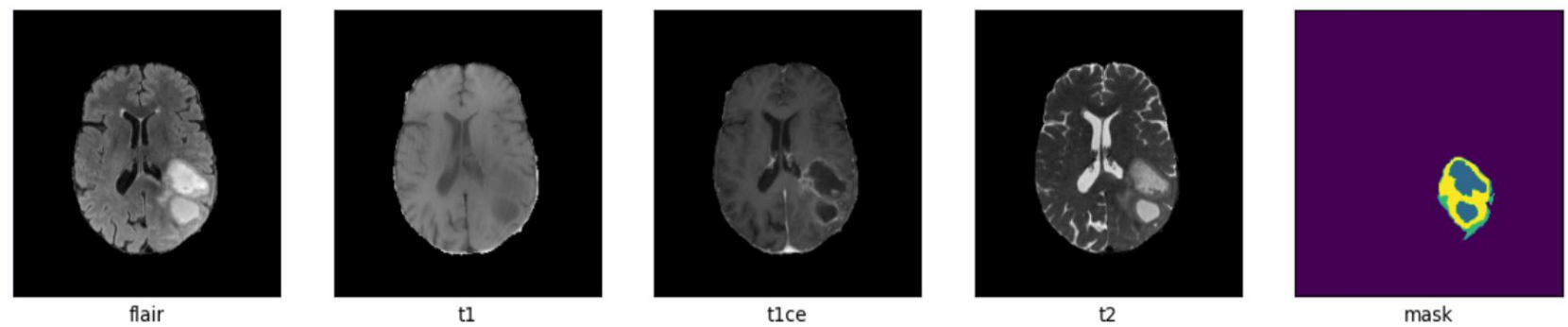



```
BraTS2021
|-- BraTS2021_00000
|   |-- BraTS2021_00000_flair.nii.gz
|   |-- BraTS2021_00000_t1.nii.gz
|   |-- BraTS2021_00000_t1ce.nii.gz
|   |-- BraTS2021_00000_t2.nii.gz
|   |-- BraTS2021_00000_seg.nii.gz
|-- BraTS2021_00002
|   |-- BraTS2021_00002_flair.nii.gz
|   |-- BraTS2021_00002_t1.nii.gz
|   |-- BraTS2021_00002_t1ce.nii.gz
|   |-- BraTS2021_00002_t2.nii.gz
```

- 1251 Samples
- 4 modalities: T1, T1C, T2, FLAIR
- A segmentation mask

ABOUT DATASETS

- Visualization of 4 modalities and segmentation mask



- Each modality complements information`

SYSTEM BLOCK DIAGRAM

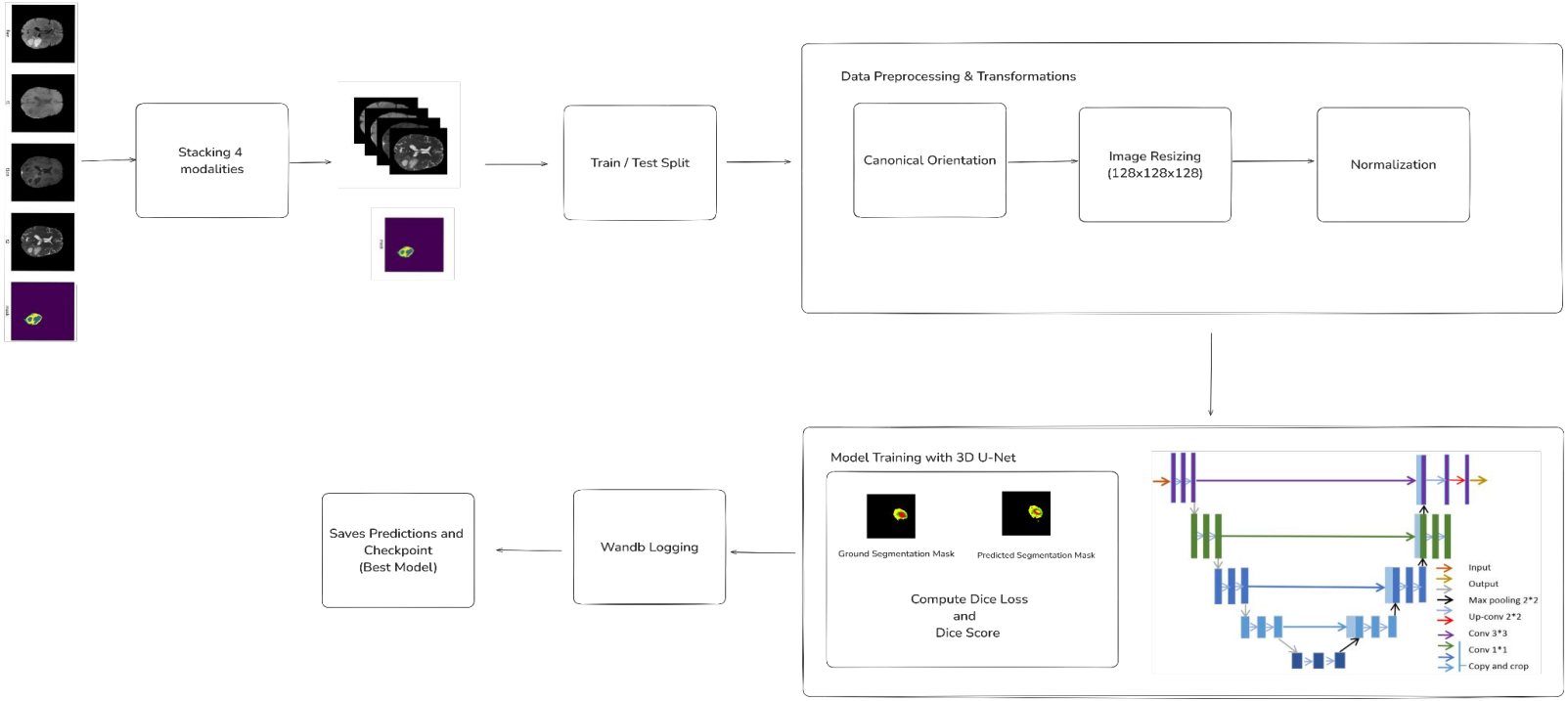


Fig: System Architecture of Baseline Model for Brain Tumor Segmentation

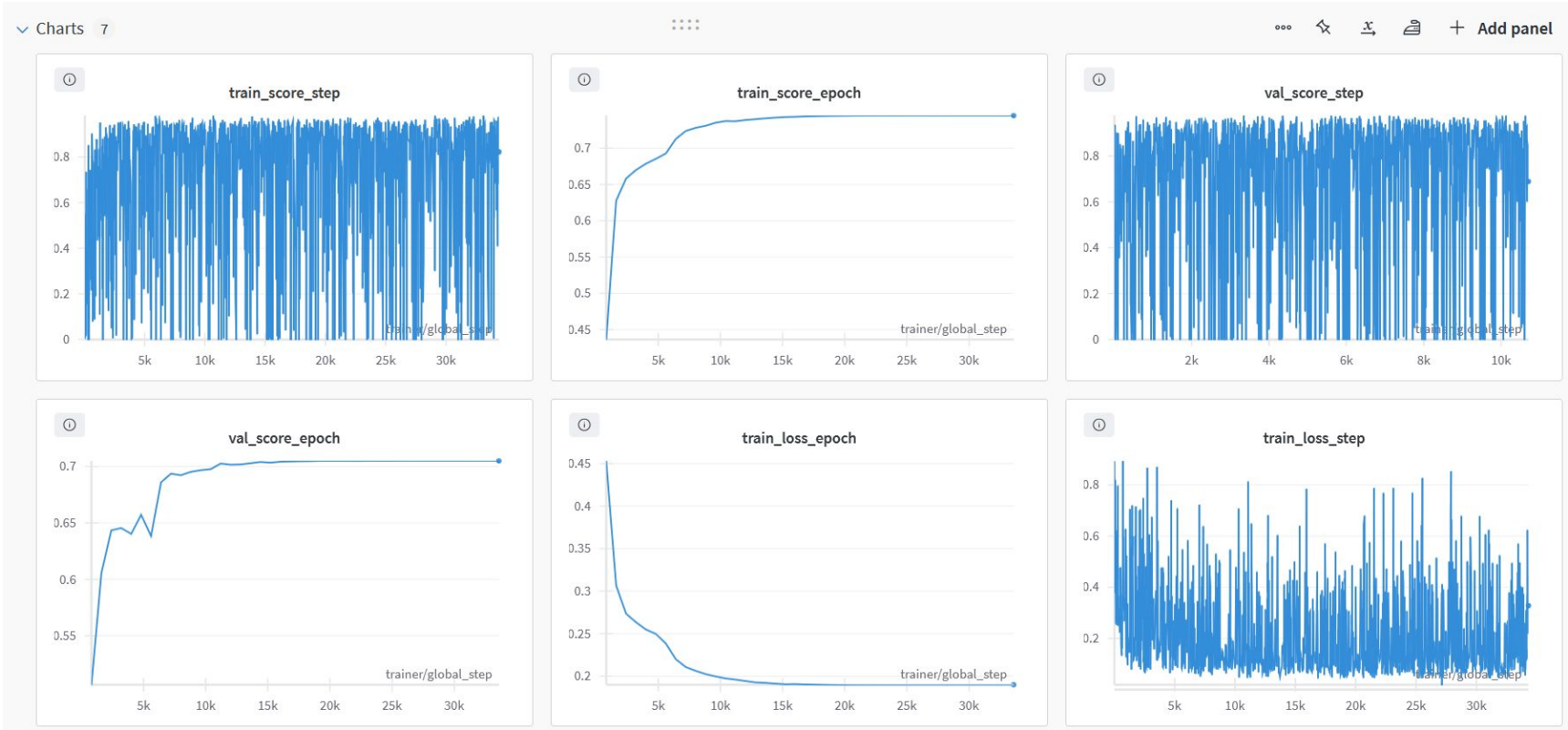
CORE AI COMPONENT ARCHITECTURE

```
model = UNet(
    spatial_dims=3,         # 3 for using 3D ConvNet and 3D Maxpooling
    in_channels=4,          # since 4 modalities
    out_channels=4,         # 4 sub-regions to segment
    channels=(16, 32, 64, 128, 256),
    strides=(2, 2, 2, 2),
).to(my_device)
```

TRAINING SETTINGS

1. **Model** => 3D U-Net
2. **Loss Function** => Dice Loss
3. **Evaluation Metric** => Dice Score
4. **Batch Size** => 1
5. **Optimizer** => Adam optimizer
6. **Learning rate** => 0.002
7. **Training Epochs** => 42 epochs (Took around 10 hours to train)
8. **Training Termination** => Early stopping monitoring val_score with patience=5

LEARNING CURVES



MODEL EVALUATION

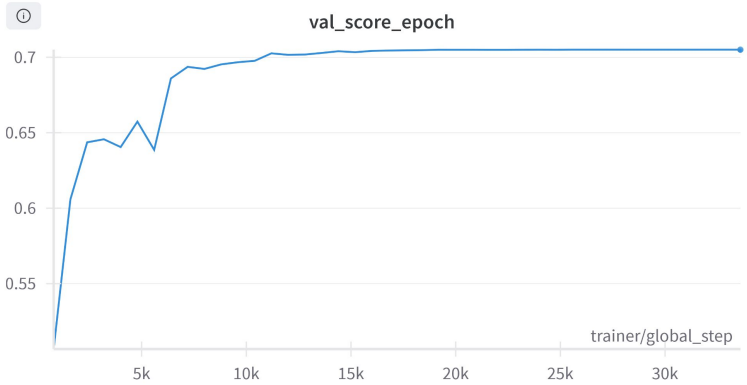
Performance from Literature on Brats-2021 Dataset using 3D-UNet

Performance (Dice Scores):

- Enhancing Tumor (ET): 0.706
- Whole Tumor (WT): 0.871
- Tumor Core (TC): 0.771

Mean Dice of ~0.78

Our Baseline Model Performance



Mean Dice of ~0.72

CONTRIBUTION AREAS

NAMEa	PROJECT CONTRIBUTION
Bijay Adhikari	● Baseline Model with Pytorch Lightning Pipeline for easy Checkpoint Loading, Early Stopping, Fine Tuning, and Logging ● Wandb Logging ● Implementing Data Augmentation Pipeline

SCOPES IN REAL WORLD

1. Faster & More Accurate Diagnosis
   Automating segmentation reduces diagnosis time and potential errors compared to manual methods.
2. Improved Treatment Planning
   Detailed tumor information aids in planning surgeries, radiation, and chemotherapy.
3. Effective Tumor Monitoring
   Segmentation allows tracking tumor changes over time to assess treatment response.
4. Integration with Healthcare Systems
   The project's output can integrate with EHRs and surgical planning software.

Key Features: Overview

- **Input Data**
  - Four MRI modalities: T1, T1c, T2, and FLAIR.
  - Data format: NIfTI (.nii.gz), organized into training and testing sets.
- **3D U-Net Architecture**
  - 3D model with 4 input channels (one for each MRI modality).
  - **Contracting path (Encoder):** Captures context.
  - **Expanding path (Decoder):** Generates segmentation mask.
- **Data Preprocessing**
  - **Modality Stacking**
  - **Canonical Orientation**
  - **Image Resizing**
  - **Normalization**

Key Features: Model Training and Evaluation

- **Training Process**
  - **Batching & Shuffling:** Efficient data loading with batch size of 1.
  - **Loss Function:** Dice Loss to compare predicted and ground truth segmentation.
  - **Optimizer:** Adam optimizer for updating parameters.

- **Validation**
  - Evaluation mode with Dice Similarity Coefficient (DSC) as the performance metric.

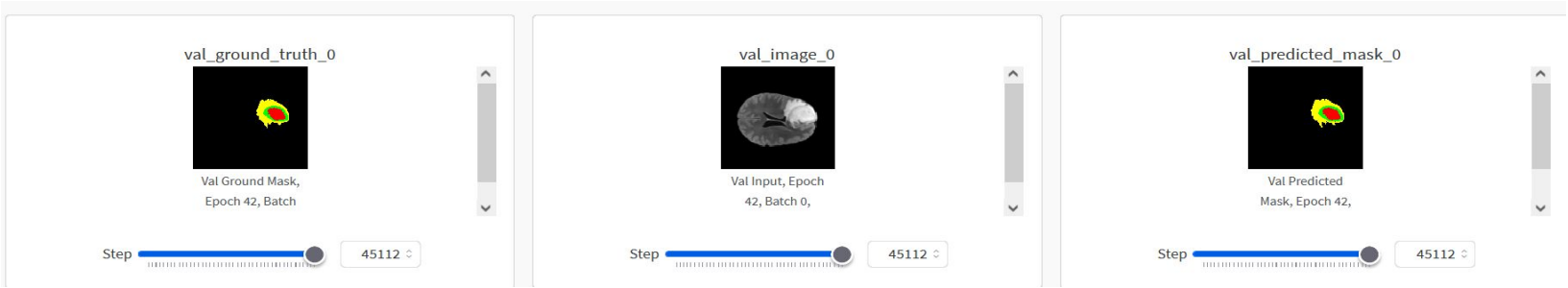
- **Checkpointing & Logging**
  - Logging metrics such as training and validation Dice Loss and Dice Score at regular intervals on the wandb
  - Logging ground truth, and prediction images to the wandb as well
  - Saving the model checkpoints

FUTURE ENHANCEMENT

- Experimenting with different **Loss Functions**
- Exploring appropriate **data augmentations** methods
- Fine tuning for **synthetic image creation**
  Other pre trained models can be fine tuned to create further instances of brain tumor segmentation dataset.
- **User friendly interface**
  A user friendly interface can be developed so that segmentation can be easily done with the help of a graphical interface.

SCREENSHOT

Visualization of Ground Truth and Predicted Mask from validation set



CONTRIBUTION AREAS

NAME	PROJECT CONTRIBUTION
Piyush Dawadi	● Created a Baseline model ● Data Visualization ● Exploratory Data Analysis